Chapter 2. Populations and Samples

Distributions, summary measures and graphical displays

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2.1 Distribution of a population

Motivation. A distribution records how a population's values are spread, as nonnegative weights summing to one.

Table 1-1 summarized a *sample*; the election result summarizes the *population*:

Candidate	J. Smith	E. Johnson	M. Lee	S. Davis	Total
Vote share	0.367	0.281	0.271	0.081	1.000

The shares are nonnegative and sum to 1; drawn as bars they form a bar graph (Figure 2-1).

Definition 2-1 (Distribution of a population)

The way the characteristic values of a population are spread, expressed by nonnegative numbers summing to 1, is called the **distribution** of the population.

A pre-election poll aims to estimate this distribution.

Illustration 2-1: Mendel's peas

Crossing RRYY with rryy (R round, r wrinkled, Y yellow, y green), the second generation shows round–yellow (C1), round–green (C2), wrinkled–yellow (C3), wrinkled–green (C4) in ratio $9:3:3:1$; endless repetition gives an infinite population with distribution

Value	C1	C2	C3	C4
Proportion	9/16	3/16	3/16	1/16

Qualitative or discrete values have distributions given by relative frequencies (or their limits).

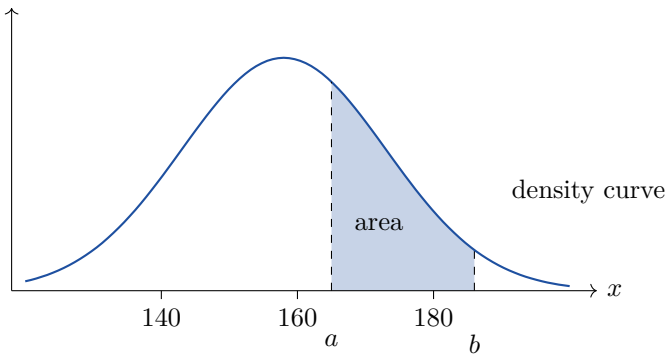
Continuous characteristics. Height, weight etc. are grouped into classes; with width 10 cm, say

Class (cm)	130–140	140–150	150–160	160–170	170–180	180–190	Total
Rel. freq.	0.05	0.1	0.3	0.3	0.2	0.05	1.00

(class $[a, b)$). Over each class draw a rectangle of *area* equal to its relative frequency (height = rel. freq./width); total area is 1. Width 5 cm gives a finer picture (Table 2-4: $0.01, 0.04, 0.04, 0.06, 0.2, 0.1, 0.2, 0.1, 0.05, 0.15, 0.03, 0.02$ for $130, 135, \ldots, 185$); a smooth curve removes the dependence on the classes.

Definition (Density curve)

A curve describing the distribution of an infinite population with a continuous characteristic is a **density curve**. The area under it over $[a, b]$ equals the limit of the relative frequency of values in $[a, b]$, so the total area between curve and horizontal axis is always 1. Even for a finite population, a density curve is a convenient approximation when the population is large.



Area over $[a, b]$ = limit of the relative frequency of values in $[a, b]$ (Figure 2-5)

For several characteristics, a **joint distribution** spreads mass 1 over multivariate values (bars summing to 1, or volume 1 under a surface).

2.2 Summary measures of a population

Motivation. A few numbers summarize location, spread and association of a numerical distribution.

2.2.1 Location: population mean, percentiles

For a finite population $\{c_1, \ldots, c_N\}$ (c_i the value of unit i), the **population mean** is

$$\mu = \frac{1}{N}(c_1 + c_2 + \cdots + c_N) = \frac{1}{N}\sum_{i=1}^{N} c_i.$$

If the distinct values are $c_1^*, \ldots, c_k^*$ with frequencies $f_1, \ldots, f_k$, then μ is the sum of (value)×(relative frequency):

Definition 2-2 (Population mean of a finite population)

$$\mu = \sum_{i=1}^{k} c_i^* \frac{f_i}{N}.$$

Example 2-1 (Number of children)

Number of children of the 16,549 thousand married Korean women aged 15+ (2020 census; 5* means "5 or more", counted as 5):

Children	0	1	2	3	4	5*
Rel. freq.	0.083	0.189	0.477	0.150	0.054	0.047

Find the population mean.
Solution.

$$\mu = 0(0.083) + 1(0.189) + 2(0.477) + 3(0.150) + 4(0.054) + 5(0.047) = 2.044.$$

μ is the **balance point** of the bar graph (Figure 2-7), or of a plate shaped like the density curve.

Definition 2-3 (Population mean of an infinite population)

$$\mu = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous case)} \end{cases}$$

Here $p(x)$ is the limiting relative frequency of the value x (discrete case) or the density curve (continuous case); p is called the **density function**.

Example 2-2 (Mean of an infinite population)

Find the population mean of the infinite population

x	0	1	2	3	4
$p(x)$	1/10	3/10	3/10	2/10	1/10

Solution.

$$\mu = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{3}{10} + 2 \cdot \tfrac{3}{10} + 3 \cdot \tfrac{2}{10} + 4 \cdot \tfrac{1}{10} = 1.9.$$

The p**th percentile** has $p\%$ of the ordered values below it; the 25th, 50th, 75th are the **first quartile** Q_1, **median** Q_2, **third quartile** Q_3. The median halves the area under the density; $\mu = Q_2$ for symmetric densities, and a right tail pulls μ above Q_2 (Figure 2-9). μ (with σ) is the key inference target.

2.2.2 Spread: MAD, IQR, variance

For a finite population, the **mean absolute deviation** and **interquartile range** are

$$\text{MAD} = \frac{1}{N}\sum_{i=1}^{N} |c_i - Q_2|, \qquad \text{IQR} = Q_3 - Q_1.$$

The standard measures are the **standard deviation** σ and **variance** σ^2:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2.$$

In terms of distinct values c_i^* with frequencies f_i:

Definition 2-4 (Variance and s.d. of a finite population)

(1) Population variance: $\sigma^2 = \sum_{i=1}^{k}(c_i^* - \mu)^2 \frac{f_i}{N}$.

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Definition 2-5 (Variance and s.d. of an infinite population)

For an infinite population with density function $p(x)$:

(1) Population variance:

$$\sigma^2 = \begin{cases} \displaystyle\sum_{\text{all } x} (x-\mu)^2 p(x) & \text{(discrete case)} \\ \displaystyle\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\,dx & \text{(continuous case)} \end{cases}$$

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Example 2-3 (Variance)

Find the population variance and standard deviation for Example 2-2.
Solution. With $\mu = 1.9$,

$$\begin{aligned}\sigma^2 &= (0-1.9)^2\tfrac{1}{10} + (1-1.9)^2\tfrac{3}{10} + (2-1.9)^2\tfrac{3}{10} + (3-1.9)^2\tfrac{2}{10} + (4-1.9)^2\tfrac{1}{10} \\ &= 1.29,\end{aligned}$$

and $\sigma = \sqrt{1.29} = 1.1358$.

2.2.3 Association: correlation

Table 2-5 gives two bivariate populations; each entry is the relative frequency of the pair (characteristic 1, characteristic 2).

Population A

char. 2 \ char. 1	0	1	2	Total
0	0.1	0.2	0.1	0.4
1	0.0	0.2	0.1	0.3
2	0.0	0.0	0.3	0.3
Total	0.1	0.4	0.5	1.0

Population B

char. 2 \ char. 1	0	1	2	Total
0	0.0	0.0	0.4	0.4
1	0.0	0.2	0.1	0.3
2	0.1	0.2	0.0	0.3
Total	0.1	0.4	0.5	1.0

The bottom margin gives the distribution of characteristic 1, the right margin that of characteristic 2. In both populations

$$\mu_1 = 1.4,\ \sigma_1 = 0.6633, \qquad \mu_2 = 0.9,\ \sigma_2 = 0.8307,$$

yet A's mass rises and B's falls, so a measure of association is needed. For a bivariate finite population $(c_{11}, c_{21}), \ldots, (c_{1N}, c_{2N})$ the **correlation coefficient** (Galton, Pearson) is

$$\rho = \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1}\Big)\Big(\frac{c_{2i}-\mu_2}{\sigma_2}\Big).$$

Proposition (Range of the correlation coefficient)

$-1 \le \rho \le 1$, because

$$0 \le \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1} - \rho\,\frac{c_{2i}-\mu_2}{\sigma_2}\Big)^2 = 1-\rho^2.$$

Proof. With $u_i = (c_{1i}-\mu_1)/\sigma_1$, $v_i = (c_{2i}-\mu_2)/\sigma_2$,

$$\frac{1}{N}\sum u_i^2 = 1, \qquad \frac{1}{N}\sum v_i^2 = 1, \qquad \frac{1}{N}\sum u_i v_i = \rho,$$

so

$$\frac{1}{N}\sum(u_i-\rho v_i)^2=\frac{1}{N}\sum u_i^2-2\rho\,\frac{1}{N}\sum u_iv_i+\rho^2\,\frac{1}{N}\sum v_i^2=1-2\rho^2+\rho^2=1-\rho^2.$$

and $\rho^2\le 1$. ■

Grouping distinct values c_{1i}^* $(i\le k)$ of characteristic 1 and c_{2j}^* $(j\le l)$ of characteristic 2, with f_{ij} the frequency of (c_{1i}^*,c_{2j}^*):

Definition 2-6 (Correlation of a finite population)

$$\rho=\sum_{i=1}^{k}\sum_{j=1}^{l}\left(\frac{c_{1i}^*-\mu_1}{\sigma_1}\right)\left(\frac{c_{2j}^*-\mu_2}{\sigma_2}\right)\frac{f_{ij}}{N}=\frac{\sum_{i=1}^{k}\sum_{j=1}^{l}(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\,f_{ij}/N}{\sigma_1\sigma_2}.$$

Example 2-4 (Correlations of Table 2-5)

Find the population correlation coefficients of populations A and B.
Solution. For A,

$$\begin{aligned}&\sum_i\sum_j(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\frac{f_{ij}}{N}\\&\quad=(0-1.4)(0-0.9)(0.1)+(1-1.4)(0-0.9)(0.2)+\cdots+(2-1.4)(2-0.9)(0.3)\\&\quad=0.34,\end{aligned}$$

so $\rho=0.34/(0.6633\times 0.8307)=0.6171$. For B the numerator is -0.46 and $\rho=-0.8348$.

For an infinite bivariate population let $p(x,y)$ be the **joint density function** (limiting relative frequency, or density surface):

Definition 2-7 (Correlation of an infinite population)

$$\rho=\begin{cases}\dfrac{\sum_{\text{all }x}\sum_{\text{all }y}(x-\mu_1)(y-\mu_2)\,p(x,y)}{\sigma_1\sigma_2} & \text{(discrete case)}\\[2ex]\dfrac{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)\,p(x,y)\,dx\,dy}{\sigma_1\sigma_2} & \text{(continuous case)}\end{cases}$$

The numerator is the **covariance**, $\rho\sigma_1\sigma_2$. $\rho>0$ for rising mass, $\rho<0$ for falling, and $\rho\to\pm1$ as mass concentrates on a line. Numbers describing a population (mean, variance, covariance, correlation) are **parameters**.

2.3 Graphical display of samples

Motivation. Tables and graphs of a sample give a first picture of the population's distribution.

2.3.1 Qualitative or few-valued data

A **frequency table** lists each value with its **frequency** (count) and **relative frequency** (count / sample size). It is drawn as a **bar graph** (bar heights proportional to relative frequencies) or a **pie graph** (sector areas, i.e. central angles, proportional to relative frequencies).

Example 2-5 (Bar and pie graphs)

Draw a bar graph and a pie graph of the sample in Table 1-1.
Solution. The bar graph has heights $0.42, 0.37, 0.15, 0.06$. The central angles of the pie are

$$360^\circ \times 0.42 = 151.2^\circ, \quad 360^\circ \times 0.37 = 133.2^\circ, \quad 360^\circ \times 0.15 = 54^\circ, \quad 360^\circ \times 0.06 = 21.6^\circ.$$

2.3.2 Frequency tables, histograms, polygons

For continuous or many-valued data, divide the range into classes: not too sparse, not so wide as to hide the shape, with no observation on a boundary.

Constructing a frequency table

(1) Range = (largest value) − (smallest value).
(2) Choose about 5–20 classes; class width = range ÷ (number of classes), rounded *up* to the smallest unit of the data.
(3) Start the first class at (smallest value) $-\frac{1}{2}\times$(smallest unit), and build the classes with the width from (2).
(4) Count the (relative) frequency of each class.

Example 2-6 (Heights of 40 people)

Heights (cm) of 40 randomly selected people:

183	168	165	158	180	176	168	160	160	150
180	183	170	170	185	170	165	170	168	154
152	173	173	165	163	163	160	178	138	171
163	168	165	165	178	170	168	183	145	174

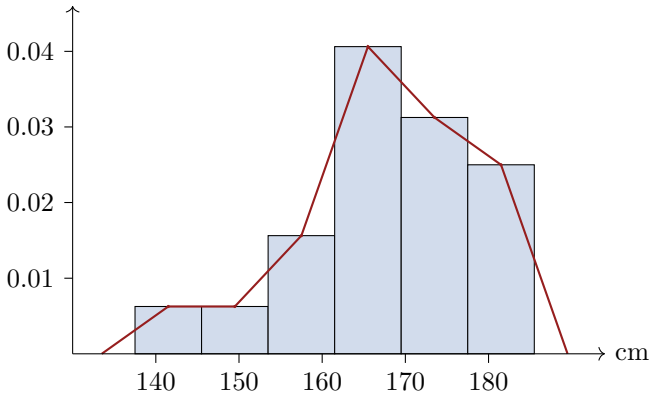
Make a frequency table.
Solution. Max 185, min 138, range 47. With 6 classes, $47/6 = 7.83\ldots$, rounded up to width 8. The first class starts at $138 - \frac{1}{2} \times 1 = 137.5$.

Class (cm)	Freq.	Rel. freq.
$[137.5, 145.5)$	2	0.050
$[145.5, 153.5)$	2	0.050
$[153.5, 161.5)$	5	0.125
$[161.5, 169.5)$	13	0.325
$[169.5, 177.5)$	10	0.250
$[177.5, 185.5)$	8	0.200
Total	40	1

A **histogram** marks the class intervals on the horizontal axis and draws over each class a rectangle with

$$\text{area} = \text{relative frequency}, \qquad \text{i.e.} \qquad \text{height} = \frac{\text{relative frequency}}{\text{class width}}.$$

Total area is 1, so the histogram estimates the density curve. Widths may vary: narrow where data are dense, wide where sparse (Figure 2-13 splits the 4th class in two). A **frequency polygon** joins the midpoints of the rectangle tops, ending on the axis half a width beyond the end classes; its area is also 1.



Histogram (height = rel. freq./8) and frequency polygon for Example 2-6 (Figures 2-12, 2-14)

2.3.3 Stem-and-leaf display

Leading digits form the **stems**; the last digit of each observation is a sorted **leaf** beside its stem. For Example 2-6:

18	0 0 3 3 3 5
17	0 0 0 0 0 1 3 3 4 6 8 8
16	0 0 0 3 3 3 5 5 5 5 5 8 8 8 8 8
15	0 2 4 8
14	5
13	8

This **stem-and-leaf display** is a histogram rotated by 90° with constant leaf width (here 10; split stems 14^-, 14^+ give 5). It keeps the data values but is impractical for large samples.

2.3.4 Bivariate samples

A **contingency table** lists the (relative) frequency of each pair of values; e.g. favourite subject of 400 students:

	Korean	English	Math	Total
Male	0.18	0.24	0.18	0.60
Female	0.14	0.16	0.10	0.40
Total	0.32	0.40	0.28	1.00

For continuous characteristics, class both ranges; the **bivariate histogram** erects over each cell a box of volume equal to its relative frequency.

Example 2-7 (Height and weight)

Adding weight to the heights of Example 2-6 (height cm, weight kg):

(183, 82)	(168, 52)	(165, 52)	(158, 55)	(180, 93)
(176, 57)	(168, 60)	(160, 48)	(160, 50)	(150, 49)
(180, 79)	(183, 73)	(170, 60)	(170, 66)	(185, 74)
(170, 82)	(165, 57)	(170, 54)	(168, 57)	(154, 55)
(152, 50)	(173, 63)	(173, 60)	(165, 49)	(163, 63)
(163, 57)	(160, 57)	(178, 77)	(138, 54)	(171, 69)
(163, 50)	(168, 57)	(165, 54)	(165, 59)	(178, 79)
(170, 54)	(168, 79)	(183, 70)	(145, 37)	(174, 65)

Make a contingency table and a bivariate histogram.
Solution. Using the 6 height classes of Example 2-6 and 6 weight classes of width 10 starting at 36.5:

Height \ Weight	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5	Total
137.5–145.5	0.025	0.025	0	0	0	0	0.050
145.5–153.5	0	0.050	0	0	0	0	0.050
153.5–161.5	0	0.100	0.025	0	0	0	0.125
161.5–169.5	0	0.125	0.175	0	0.025	0	0.325
169.5–177.5	0	0.050	0.150	0.025	0.025	0	0.250
177.5–185.5	0	0	0	0.075	0.100	0.025	0.200
Total	0.025	0.350	0.350	0.100	0.150	0.025	1.000

The bivariate histogram erects over each cell a box of volume equal to the entry (Figure 2-17).

2.3.5 Scatter plots and regressograms

A **scatter plot** plots each pair as a point. A **regressogram** classes one characteristic and plots the class means of the other; for Example 2-7 by weight class:

Weight class	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5
Mean height	145.0	159.9	168.3	180.5	176.2	180.0

As a step function this is the regressogram (Figure 2-19), usually more effective than the scatter plot for analysing the relationship.

2.4 Summary measures of a sample

Motivation. Unknown parameters are estimated by the corresponding sample quantities.

2.4.1 Location

Let the sample values be $x_1, \dots, x_n$.

Definition 2-8 (Sample mean)

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Sample percentiles, quartiles and median $\hat{Q}_1, \hat{Q}_2, \hat{Q}_3$ are defined as for the population. Since a few outlying values strongly affect $\bar{x}$, the $100p\%$ **trimmed mean** $\bar{x}_p$ drops the smallest and largest $100p\%$ and averages the remaining $100(1-2p)\%$.

Example 2-8 (Sample mean, quartiles, trimmed mean)

For the data of Example 2-6, find the sample mean, the sample quartiles and the 10% trimmed mean.

Solution. $\bar{x} = \frac{1}{40}(183 + 168 + \cdots + 174) = \frac{6698}{40} = 167.45$. From the stem-and-leaf display (midpoints of the 10th/11th, 20th/21st, 30th/31st values):

$$\hat{Q}_1 = 163, \qquad \hat{Q}_2 = 168, \qquad \hat{Q}_3 = 173.5.$$

The 10% trimmed mean drops 4 values at each end (138, 145, 150, 152 and 183, 183, 183, 185) and averages the middle 32:

$$\bar{x}_{0.1} = \frac{1}{32}(180 + 180 + \cdots + 154) = \frac{5379}{32} = 168.0938.$$

2.4.2 Spread

The **sample interquartile range** is $\widehat{\mathrm{IQR}} = \hat{Q}_3 - \hat{Q}_1$. Replacing μ by $\bar{x}$ in $\frac{1}{n}\sum(x_i - \mu)^2$ *underestimates* σ^2:

Proposition (Centring at the sample mean)

For any number μ,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2 - (\bar{x} - \mu)^2.$$

Proof. [Supplement] Square $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$:

$$\sum_{i=1}^{n}(x_i - \mu)^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + 2(\bar{x} - \mu)\sum_{i=1}^{n}(x_i - \bar{x}) + n(\bar{x} - \mu)^2.$$

The middle sum is $\sum x_i - n\bar{x} = 0$; divide by n. ∎

Dividing by $n - 1$ instead corrects this, giving the **sample variance**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Definition 2-9 (Sample standard deviation)

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Example 2-9 (Sample IQR and s.d.)

For the data of Example 2-6, find the sample interquartile range and the sample standard deviation.

Solution. From Example 2-8, $\widehat{\mathrm{IQR}} = 173.5 - 163 = 10.5$. Next

$$s^2 = \frac{1}{40-1}\{(183 - 167.45)^2 + (168 - 167.45)^2 + \cdots + (174 - 167.45)^2\} = 108.6128,$$

so $s = \sqrt{108.6128} = 10.4218$.

2.4.3 Bivariate samples

For a bivariate sample $(x_1, y_1), \ldots, (x_n, y_n)$:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad s_1 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}; \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \quad s_2 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

Definition 2-10 (Sample correlation coefficient)

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_1}\right)\left(\frac{y_i - \bar{y}}{s_2}\right) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

(The two expressions agree since $(n-1)s_1 s_2 = \sqrt{\sum (x_i - \bar{x})^2}\sqrt{\sum (y_i - \bar{y})^2}$.) The **sample covariance**, estimating the population covariance, is

$$r s_1 s_2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

Example 2-10 (Sample correlation)

Find the sample correlation coefficient for the data of Example 2-7.

Solution. $\bar{x} = 167.45$, $s_1 = 10.42$ (Examples 2-8, 2-9); $\bar{y} = 61.45$, $s_2 = 11.94$. The sample covariance is

$$r s_1 s_2 = \frac{1}{40-1}\{(183 - 167.45)(82 - 61.45) + \cdots + (174 - 167.45)(65 - 61.45)\} = 90.59,$$

so $r = 90.59/(10.42 \times 11.94) = 0.73$.

Sample quantities are **statistics**; one used to estimate a parameter is an **estimator**.